

Reduced thermal tolerance in a coral carrying CRISPR-induced mutations in the gene for a heat-shock transcription factor

Phillip A. Cleves^{a,1} , Amanda I. Tinoco^a, Jacob Bradford^b , Dimitri Perrin^b, Line K. Bay^{c,2}, and John R. Pringle^{a,2}

^aDepartment of Genetics, Stanford University School of Medicine, Stanford, CA 94305; ^bScience and Engineering Faculty, Queensland University of Technology, Brisbane, QLD 4001, Australia; and ^cAustralian Institute of Marine Science, Townsville, QLD 4810, Australia

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Reef-building corals are keystone species that are threatened by anthropogenic stresses including climate change. To investigate corals' responses to stress and other aspects of their biology, numerous genomic and transcriptomic studies have been performed, generating many hypotheses about the roles of particular genes and molecular pathways. However, it has not generally been possible to test these hypotheses rigorously because of the lack of genetic tools for corals or closely related cnidarians. CRISPR technology seems likely to alleviate this problem. Indeed, we show here that microinjection of single-guide RNA/Cas9 ribonucleoprotein complexes into fertilized eggs of the coral *Acropora millepora* can produce a sufficiently high frequency of mutations to detect a clear phenotype in the injected generation. Based in part on experiments in a sea-anemone model system, we targeted the gene encoding Heat Shock Transcription Factor 1 (*HSF1*) and obtained larvae in which >90% of the gene copies were mutant. The mutant larvae survived well at 27 °C but died rapidly at 34 °C, a temperature that did not produce detectable mortality over the duration of the experiment in wild-type (WT) larvae or larvae injected with Cas9 alone. We conclude that HSF1 function (presumably its induction of genes in response to heat stress) plays an important protective role in corals. More broadly, we conclude that CRISPR mutagenesis in corals should allow wide-ranging and rigorous tests of gene function in both larval and adult corals.

Acropora | genome editing | heat stress

Reef-building corals are ecologically, economically, and aesthetically critical species in the tropical and subtropical oceans. However, corals have declined rapidly during recent decades due to a variety of anthropogenic stresses including the ocean warming associated with climate change (1–4). These stresses often lead to coral bleaching, the breakdown of the symbiosis between the coral animals and their dinoflagellate algae endosymbionts, whose photosynthesis typically provides shallow-water corals with most of their energy and a variety of organic metabolites. If the heat stress is severe or prolonged, death can result. Despite their importance, corals and their algal symbionts remain poorly understood at the molecular and cellular levels, which seriously handicaps efforts to predict, prevent, and/or mitigate the effects of stress.

Two major obstacles to a better understanding of coral molecular and cell biology have been the lack of experimentally tractable model systems and the lack of genetic methods to allow rigorous testing of hypotheses about gene and protein function. However, it has recently become clear that *Aiptasia* (*sensu* *Exaiptasia pallida*), a small sea anemone symbiotic with dinoflagellates like those found in corals, has great promise as a model system for studying many aspects of coral biology (5–14). Moreover, the wide applicability of the CRISPR/Cas9 technology for gene modification (15–17) offers the exciting potential to begin genetic testing of hypotheses in both *Aiptasia* and corals. Indeed, we recently reported that microinjection of single-guide RNA (sgRNA)/Cas9 ribonucleoprotein complexes into freshly fertilized eggs of the reef-building coral *Acropora millepora* could

produce the expected types of mutations (18). However, the frequency of mutations was much less than 100%, and we did not observe associated phenotypes, whether because of the incomplete mutagenesis, the particular genes chosen to target, or both.

Because of the prominent role of thermal stress in the decline of corals worldwide, a major focus of research is to understand both (i) the mechanisms that lead to heat-induced bleaching and death and (ii) those that may protect against it. Studies in many organisms, including symbiotic corals and anemones, have found an extensive burst of transcription upon heat stress; the up-regulated genes include many encoding molecular chaperones such as HSP70 and HSP90 (19–28). However, for several reasons, it has remained unclear whether this transcriptional response has a role either in producing or in protecting against bleaching and death in corals and anemones. First, similar responses are seen in both symbiotic and nonsymbiotic organisms. Indeed, in *Aiptasia*, most of the genes that are up-regulated rapidly in response to heat stress in symbiotic animals are up-regulated similarly in aposymbiotic (algae-free) animals (27). Second, the transcriptional response is transient, and the transcript levels for many genes return to near baseline before most bleaching occurs (27). Interestingly, our recent work in *Aiptasia*

Significance

Coral reefs are biodiversity hotspots of great ecological, economic, and aesthetic importance. Their global decline due to climate change and other stressors has increased the urgency of understanding the molecular bases of corals' responses to stress. Analyses of coral genomes and gene-expression patterns have identified many genes that may be important in stress resistance, but rigorous testing of their function will require the analysis of appropriate mutants. Here, we used CRISPR technology to show that mutational loss of a putative regulator of gene expression in response to heat stress indeed produced a loss of heat tolerance. Such use of CRISPR to generate mutations in corals should illuminate many aspects of coral biology and, thus, help to guide conservation efforts.

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¹Present address: Department of Embryology, Carnegie Institution for Science, Baltimore, MD 21218.

²To whom correspondence may be addressed. Email: l.bay@aims.gov.au or jpringle@stanford.edu.

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suggests that at least some of the early-responding genes are targets of the extensively studied HSF1 (29–33) and NFκB (34, 35) transcription factors (27).

These observations in *Aiptasia*, and the many studies of HSF1 function in other organisms, suggested strongly that using CRISPR/Cas9 to eliminate HSF1 function in *A. millepora* would affect its heat tolerance. Indeed, we report here that the use of improved methods for designing sgRNAs, optimizing the volume of injected sgRNA/Cas9 solution, and identifying successfully injected animals allowed us to obtain a very high frequency of mutations in the *A. millepora* *HSF1* gene and observe a clear-cut decrease in the heat tolerance of larvae carrying the mutations.

Results

Highly Effective Disruption of a Coral Heat Shock Factor 1 Gene Using CRISPR/Cas9. To ask if the burst of transcription that occurs immediately after the imposition of heat stress in symbiotic cnidarians plays a role in determining their heat tolerance, we sought to make a mutation that would reduce or eliminate this burst. Experiments in many organisms suggested that a mutation eliminating the Heat Shock Factor 1 transcription factor might achieve this goal (*Introduction*). A search of the *A. millepora* genome identified *HSF1* (*Acropora_millepora_12629*), which encodes an HSF1-like protein (*SI Appendix, Fig. S1*) and appears to be single copy in the genome (*Materials and Methods*). To make mutations in *HSF1*, we designed two sgRNAs, one targeting exon 3 (in the predicted DNA-binding domain) and another targeting exon 9 (Fig. 1A; see *Materials and Methods* and *SI Appendix, Table S1*). These target sites were chosen to minimize the chances of off-target effects, avoid possible complications from alternative transcription start sites, be early enough in the coding sequence that frameshifts would be likely to create non-functional products, and (in the case of the exon 3 site) potentially disrupt a functionally important region of the protein.

These sgRNAs were used to induce mutations in *A. millepora* larvae during the November 2018 spawning of this species in northeastern Australia. The three nights of spawning were treated as three independent replicate experiments. Each night, we injected multiple batches of freshly fertilized zygotes with a solution containing a fluorescent marker (*SI Appendix, Fig. S2*) together with either Cas9 protein alone or a mixture of sgRNA1/Cas9 and sgRNA2/Cas9 ribonucleoprotein complexes (*Materials and Methods*). For each experiment, we attempted to inject 100–300 individuals for each condition (Table 1). Most (68–90%) of these individuals survived and appeared to develop normally for 12 h (Table 1), at which point we sorted them according to whether the fluorescent injection marker was visible (*SI Appendix, Fig. S2*); 69–90% of the surviving larvae were positive for the marker (Table 1).

At 6 d postfertilization in each experiment, we took 10–12 larvae that had been successfully injected with the sgRNA/Cas9 complexes and determined their genotypes at both the sgRNA1 and sgRNA2 sites by amplicon sequencing; 10–16 PCR clones were sequenced for each site in each larva (*SI Appendix, Table S2*). Five larvae from each experiment that had been successfully injected with Cas9 alone were analyzed similarly as a control (five PCR clones sequenced per animal). Remarkably, nearly all (29/32) of the sgRNA/Cas9-injected animals yielded $\geq 70\%$ mutant sequences at one or both sites, while 75% (24/32) yielded exclusively mutant sequences at one or both sites (Fig. 1B and *SI Appendix, Table S2*). Overall, it appeared that $\sim 99\%$ of the *HSF1* copies were mutant (at one or both sites) in experiments 1 and 3 and $\geq 80\%$ in experiment 2 (Fig. 1B, *Upper*, bottom line; *SI Appendix, Table S2*). In contrast, no mutant sequences were detected in any of the control animals (Fig. 1B).

Although there was a high overall frequency of mutations, their patterns were not the simple ones expected if most had been induced during the first one or two cell cycles. Although we

observed one case in which only two distinct mutations were found (Fig. 1C, Larva 1, Exon 9), consistent with mutation induction during the first cell cycle, and a few others with only three or four distinct mutations (*SI Appendix, Table S2*), most larvae examined had more complicated patterns, with as many as 12 distinct mutant alleles observed at a single site (Fig. 1C and *SI Appendix, Table S2*). These data suggest that the sgRNA/Cas9 complexes remained active and continued to produce mutations in previously wild-type (WT) chromosomes, for multiple cell cycles. Nonetheless, we also observed some WT sequences (Fig. 1C and *SI Appendix, Table S2*), indicating that some copies of *HSF1* had escaped mutagenesis at one or possibly both sites. However, given the overall mutation frequencies at each site, the frequency of fully WT alleles of *HSF1* should be very low (*Discussion*). Note also that as expected, $\sim 2/3$ of the mutant alleles produced frameshift mutations, presumably guaranteeing a loss of gene-product function.

It has been shown previously that the presence of more than one double-strand break in a chromosome can result in the production of large deletions between the breakpoints during DNA repair (36, 37). To determine if such deletions were generated between the sgRNA1 and sgRNA2 sites in our injected larvae, we attempted to amplify a PCR product spanning exons 3–9 (Fig. 1A, primers 1 and 4). No product was detected with DNA from control animals injected with Cas9 protein only, or from 19 of the 32 animals injected with the sgRNA/Cas9 complexes, but an ~ 200 -bp product was amplified from the remaining 13 of those animals (*SI Appendix, Table S2*). Sequencing of these PCR products from one larva from each experiment revealed that the expected large deletions had indeed been produced (*SI Appendix, Fig. S3*).

Decreased Heat Tolerance of Coral Larvae Carrying Mutations in HSF1.

To ask if HSF1 contributes to heat tolerance in corals, we subjected uninjected, Cas9-injected, and sgRNA/Cas9-injected larvae to heat stress at 34 °C for 48 h. Both the uninjected and Cas9-injected larvae (no *HSF1* mutations) showed little mortality under these conditions (Fig. 2A, *Left*; B; and C). Similarly, we saw little mortality when sgRNA/Cas9-injected larvae were held at 27 °C (Fig. 2D). These larvae also did not obviously differ from WT larvae in appearance or swimming behavior, indicating that loss of HSF1 produces no major disruption of development to this stage. In contrast, nearly two-thirds of the sgRNA/Cas9-injected larvae died during the 48-h incubation (Fig. 2E). Death was often associated with rupture of the larvae and leakage of yolk and other contents, which began early during the period of heat stress (Fig. 2A, *Right*). Taken together with the evidence that most copies of *HSF1* were mutant in the sgRNA/Cas9-injected animals (Fig. 1B), the data indicate that HSF1 function is required for the normal heat tolerance of coral larvae.

To ask if the animals surviving until 48 h (Fig. 2E) might be mainly ones with fewer mutant copies of *HSF1*, we sequenced (as in Fig. 1 and *SI Appendix, Table S2*) PCR clones amplified from the DNA of five survivors in each experiment. Indeed, the fraction of WT sequences was higher than in the total population of mutagenized animals (*SI Appendix, Table S3*; cf. *SI Appendix, Table S2*), particularly in Experiment 2, where four of the five survivors contained largely WT sequences.

Discussion

CRISPR/Cas9 as a Tool for Generating Loss-of-Function Mutations in Corals. Reverse-genetic approaches have been enormously powerful in elucidating gene function in many organisms, and their range has been greatly extended by the CRISPR/Cas9 genome-editing technology (15, 38–41). Recently, we reported that microinjection of sgRNA/Cas9 complexes into coral zygotes could induce targeted mutations, but we did not observe associated phenotypes, probably at least in part because the mutation

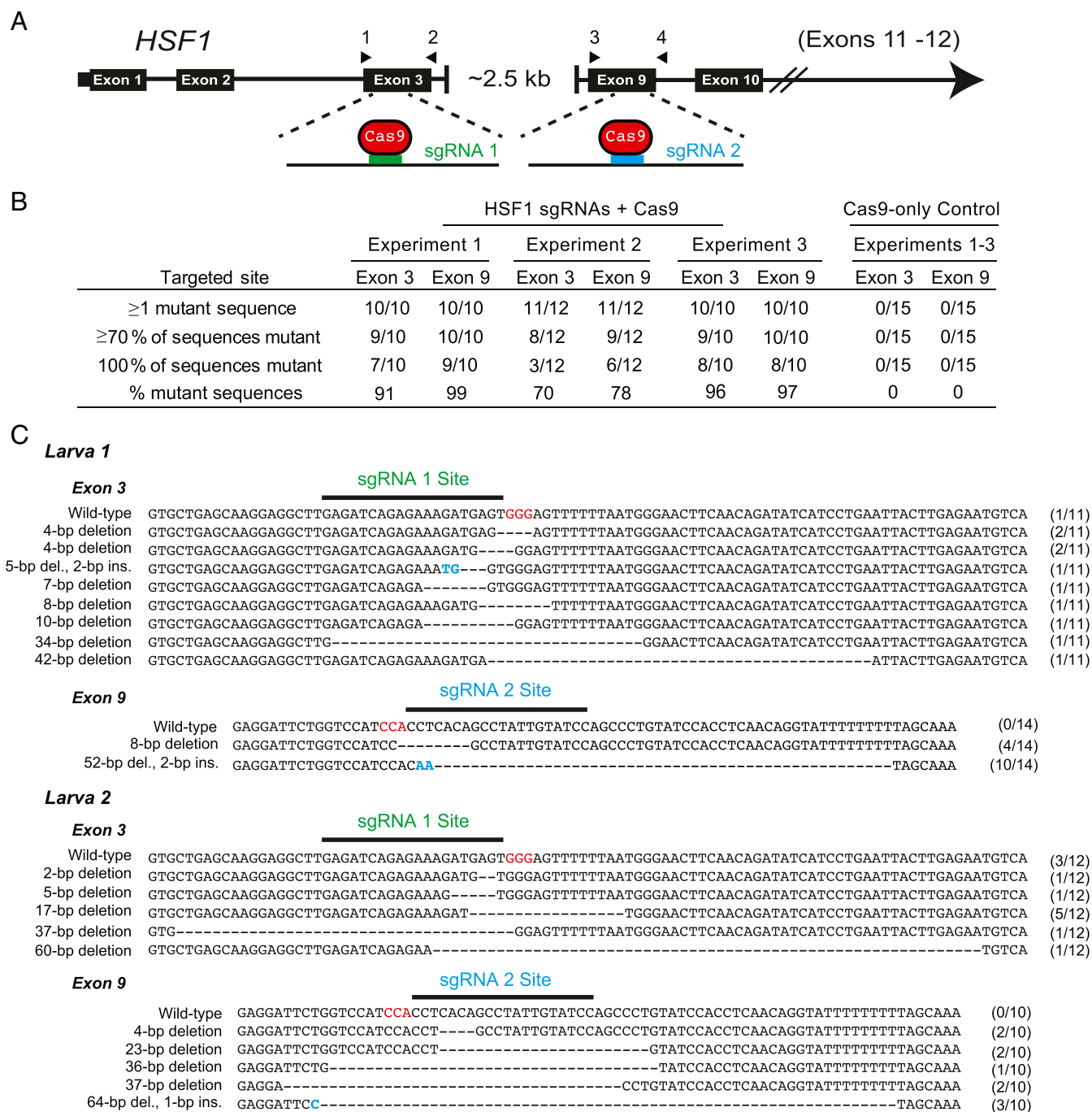


Fig. 1. Highly effective disruption of *HSF1* by CRISPR/Cas9 in *A. millepora* larvae. (A) Structure of *HSF1* and locations of the targeted sites. Five of the 12 exons are shown; the distance between exons 3 and 9 is ~2.5 kb. Approximate locations of the sites targeted by sgRNAs 1 and 2 are shown (see *SI Appendix, Table S1*, for sequences), as are the locations of the primers (arrowheads 1–4) used to amplify genomic DNA for sequencing to test for the presence of induced mutations. (B) Fractions of larvae in which ≥1, ≥70%, or 100% of the PCR clones sequenced (10–16 per animal) showed mutations and overall percentages of mutant sequences observed (see *SI Appendix, Table S2* for details). Experiments 1–3 represent the three nights on which spawning occurred and freshly fertilized eggs were injected. For control animals injected with Cas9 only (no sgRNA), data are combined for the five animals injected on each night of spawning. (C) Patterns of the mutations seen in two individual larvae. Black bars, positions of the sgRNA target sites; red nucleotides, Protospacer-adjacent-motif (PAM) sites. Dashes, nucleotides deleted in the mutant alleles; blue letters, inserted/replaced nucleotides.

frequencies were not high enough (18). In the present study, we made several changes to try to increase these frequencies. First, we used an updated protocol for sgRNA design (42–44) that is intended to increase the efficacy and specificity of the sgRNAs by identifying sequences with properties such as a low propensity for inappropriate secondary-structure formation (thus promoting efficient Cas9 binding) and no predicted potential to recognize sequences elsewhere in the genome (*Materials and Methods*). Second, we followed the previously used strategy of injecting more than one sgRNA targeting the gene of interest (37, 42, 45). If mutagenesis at each of two well-separated sites is independent, then even a modest efficiency of mutation induction at each site (e.g., 70%) would result in a low percentage of alleles with WT sequences at both sites (e.g., 30% × 30% = 9%). This stratagem

Table 1. Numbers of injected and surviving *A. millepora* zygotes

Injected material*	Experiment 1		Experiment 2		Experiment 3	
	Cas9 alone	sgRNA/Cas9	Cas9 alone	sgRNA/Cas9	Cas9 alone	sgRNA/Cas9
Nos. of individuals for which injection was attempted	100	200	200	200	300	300
Nos. of individuals surviving until 12 h postfertilization [†]	68	180	147	148	239	205
Nos. of survivors that had been injected successfully [‡]	52	125	117	133	190	154

During three nights of spawning (November 27, 28, and 30, 2018), eggs and sperm were mixed at approximately hourly intervals from 2200 to 0300, and injections of the freshly fertilized eggs were attempted within the first hour after each mixing.

*For each zygote, either Cas9 alone or a mixture of the two sgRNA/Cas9 complexes was injected, as indicated (*Materials and Methods*).

[†]As judged by the numbers of larvae that were seemingly intact at the time of observation.

[‡]As judged by the Alexa Fluor 488-dextran injection marker (*Materials and Methods*).

is particularly important given that only ~2/3 of the small deletions and insertions produced by CRISPR/Cas9 are expected to produce frameshifts that truncate translation and thus prevent the formation of a functional protein. As many other small deletions or insertions may be tolerated by the protein of interest, it may also be helpful to locate one or both target sites within functionally important regions (e.g., our targeting of one site in the presumed DNA-binding region of *A. millepora* HSF1). Finally, as an injection indicator, we replaced the previously used phenol red (whose color dissipates quickly after injection) with a high molecular weight fluorescent marker and performed the injections using a fluorescence microscope. This approach allowed both better monitoring and control of the injected volume and reliable identification of the successfully injected larvae after 12 h of development. Although we cannot tell how much each of these changes may have contributed individually to the improved mutation frequencies observed here, it does seem clear that the revised protocol, as a whole, now yields frequencies high enough to allow effective analysis of gene function in *A. millepora* and presumably in other corals as well.

The Role of HSF1 in Coral Heat Tolerance and Adaptation. Research in many organisms has revealed a rapid transcriptional response to heat stress that is highly conserved and helps to maintain cellular homeostasis (19–21). Among the up-regulated genes are those encoding members of the heat-shock-protein (HSP) family, molecular chaperones that play a major role in correcting protein misfolding. This transcriptional response is controlled in part by the transcription factor HSF1 (30, 31, 33), so that reducing HSF1 function can have dramatic effects on organismal heat tolerances (29, 32). Given this background, it is not surprising that many studies in corals and other cnidarians have found induction of genes encoding HSPs and HSF1 during heat stress (23–27, 46). Here, we have shown that ablation of HSF1 function in *A. millepora* by CRISPR/Cas9-induced mutations leads to rapid death of larvae under heat stress. However, given the presumed multiplicity of HSF1 targets, it remains unclear which of them are most important for protection against such stress.

Because heat-induced bleaching and death are important factors in the worldwide deterioration of coral reefs (1–4), it is of great interest to identify the genes that might evolve to help corals adapt to a changing climate. Given its central role in the heat-stress response, *HSF1* is an intriguing candidate, and it seems possible that variations in its levels of baseline expression and/or up-regulation under heat stress could underlie organismal differences in heat tolerance. However, it is also not clear what the overall fitness consequences of such differences in expression might be over a range of environmental conditions. These issues might be illuminated by asking if there are differences in *HSF1* expression between organisms with natural differences in heat tolerance, such as individuals of different species (47) or populations from different environments (48).

Future Prospects for Coral Genetics. In this study, we knocked out an individual gene by microinjection of sgRNA/Cas9 complexes into zygotes and examined the resulting phenotype in larvae. This same approach should allow critical evaluation of the functions of many other genes that may be involved in determining stress tolerance, the establishment and maintenance of symbiosis, or aspects of basic cellular and developmental biology. However, maximizing the power of genetic methods in coral biology will also require an expansion of the approaches used, presumably including at least the following.

First, there will be cases in which two (or several) paralogous genes encode products that function redundantly, so that knocking out just one gene may not yield an informative phenotype. Fortunately, the mutation frequencies demonstrated here appear to be high enough that it should be possible to knock out two (or possibly more) such genes simultaneously by injecting two or more sgRNAs for each, producing a sufficient loss of overall gene function that the relevant phenotype will be manifested. This strategy has been used successfully in several studies in mice to produce double knockouts without a need for crosses (42, 49). In addition, it has long been recognized that the examination of “synthetic effects,” in which a strain carrying mutations in each of two nonparalogous genes has a much more severe phenotype than either of the corresponding single mutants, is a very powerful approach to the identification of physical and/or functional interactions between gene products (50). It may be feasible to apply this approach in corals if the required double mutants can indeed be generated directly in the injected generation without a need for difficult and time-consuming genetic crosses.

Second, some interesting and ecologically important traits, such as biomineralization, cannot be studied in coral larvae. It is also possible that the genetic control of stress resistance, or other traits, may differ in some way(s) between larval, juvenile, and adult corals. Indeed, there is conflicting evidence as to whether the specificity of symbiont uptake and/or maintenance differs at different stages of the life cycle (7, 51–55). Fortunately, for a variety of coral species (including *A. millepora*), methods for inducing metamorphosis, settlement, and nascent colony formation are available (56, 57), and there seems to be no reason that these methods cannot be applied to CRISPR-mutagenized as well as WT larvae, so long as appropriately contained rearing facilities are available.

Third, microinjection is a rather labor-intensive method that allows investigation of only one or a few genes at a time, a problem compounded by the fact that zygotes for injection are typically available only seasonally when the coral of interest spawns. Thus, there will undoubtedly be situations in which a higher-throughput (even if less definitive) method will be useful for quickly testing a group of genes for relevance to the trait of interest. For example, the specific roles of HSF1 in promoting heat tolerance are likely to be mediated by subsets of its many predicted targets (27, 29, 31, 32), and deciding which of these genes to study in detail is a challenge. It seems possible that using

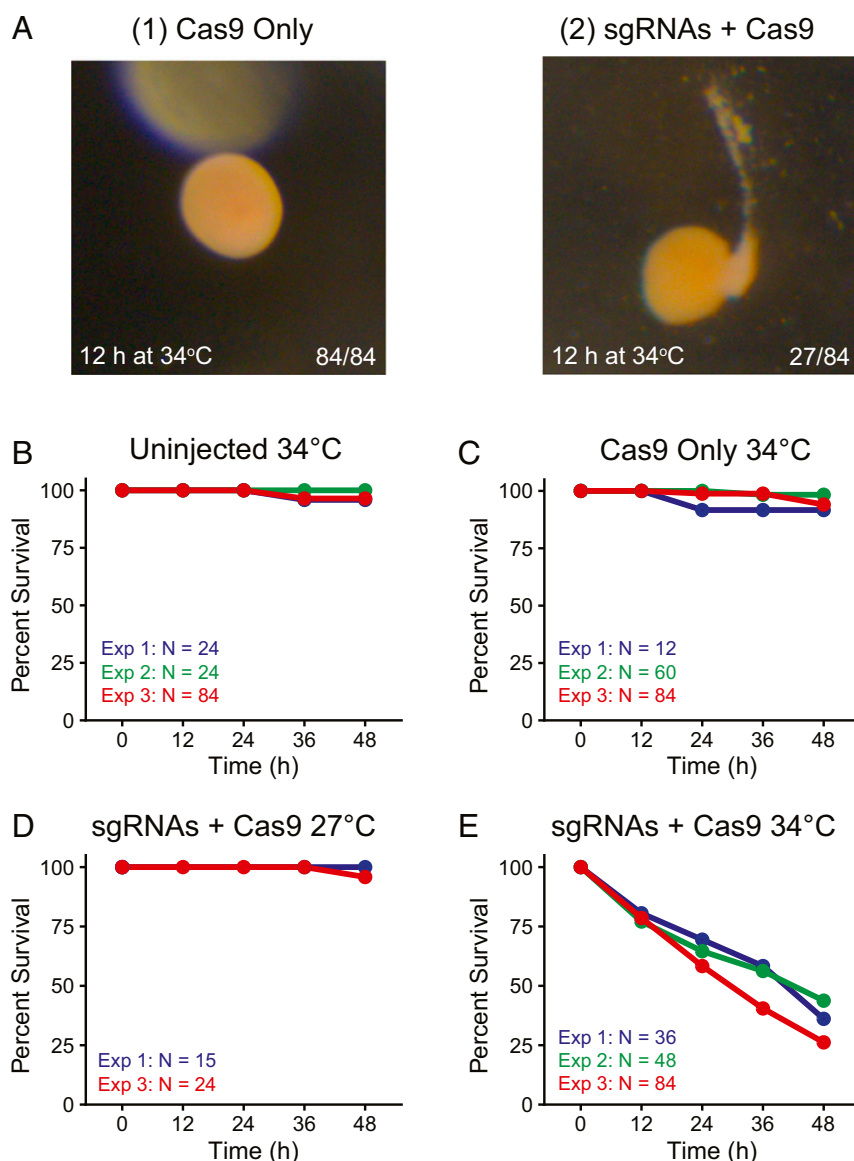


Fig. 2. Reduced heat tolerance in *A. millepora* larvae harboring mutations in *HSF1*. (A) Larvae from the populations in Experiment 3 (Fig. 1B) that had been injected with Cas9 alone or with the sgRNA/Cas9 complexes (84 larvae in each case) were examined using a stereomicroscope after incubation for 12 h at 34 °C. Numbers indicate the fraction of larvae in each population that resembled the individual shown. (B and C) Survival curves (percentages of surviving animals) for uninjected (B) and Cas9-injected (C) larvae during incubation at 34 °C. (D and E) Survival curves for sgRNA/Cas9-injected larvae during incubation at 27 °C (D) or 34 °C (E). The numbers of individual larvae monitored in each experiment are indicated. No larvae were incubated at 27 °C in Experiment 2.

electroporation or chemical transfection methods to deliver sgRNA/Cas9 complexes or gene-knockdown reagents (e.g., morpholinos, shRNAs, or CRISPRi; refs. 58–60) into large groups of zygotes would allow a rapid screening for relevant phenotypes that could then be followed up with more detailed analyses of the individual genes that seem most interesting.

Fourth, although the phenotypes resulting from gene overexpression are typically harder to interpret and more prone to artifact than those resulting from gene knockout or knockdown, they can sometimes be informative. For example, the heat sensitivity of *HSF1*-deficient larvae suggests that increasing *HSF1* expression might increase heat tolerance. However, this prediction is far from certain, given that *HSF1* expression is already so rapidly and massively up-regulated at the beginning of heat stress (27, 31, 33) and that activity of the *HSF1* protein is also regulated in vivo (61). Thus, it would be of interest to examine the

effects on heat tolerance of overexpressing *HSF1* by microinjection of in vitro synthesized mRNA, a plasmid containing *HSF1* under control of a high-expression constitutive promoter, or the reagents for introducing such a construct into the genome by CRISPR-mediated insertion (62).

In conclusion, it now seems feasible to apply a variety of genetic approaches to corals and related symbiotic cnidarians, which should lead to a dramatic improvement in our understanding of their molecular, cellular, and developmental biology. This, in turn, should provide a more solid foundation for important conservation goals such as identifying potentially more resilient populations and developing and assessing possible ameliorating actions. Given the present and future dire threats to coral reefs from anthropogenic stresses including climate change, there may also be a temptation to try to engineer genetically “improved” corals (e.g., strains more tolerant of high temperatures) and

introduce them into the wild. However, such attempts should be approached with extreme caution, given that our understanding of the underlying biological mechanisms remains so primitive that we cannot yet predict with confidence the full range of positive or negative consequences of any such genetic alteration, even for the individual organisms, much less at the ecosystem level.

Materials and Methods

Animal Collection, Spawning, and Larvae Husbandry. Individual colonies of *A. millepora* were collected in mid-November 2018 from the Great Barrier Reef, transferred to the National Sea Simulator at the Australian Institute of Marine Science, and held in outdoor flow-through tanks under ambient light and temperature conditions. The colonies spawned on November 27, 28, and 30 at ~2100. During each night of spawning, egg and sperm bundles from several colonies were collected and washed gently in 120- μ m mesh sieves to break up the bundles and separate the eggs from most of the sperm. For the microinjection experiments, eggs from each of the spawned colonies were kept separate until fertilization. Fertilizations used pooled sperm from several individuals at a final concentration of $\sim 10^6$ mL⁻¹ and were staged at 1-h intervals for ~5 h. This staggering provided time for microinjection of each batch of zygotes prior to their first cleavage, which typically occurs by 60–90 min postfertilization. Fertilized eggs were incubated at densities of <1 larva·mL⁻¹ in filtered seawater to allow for further development and experimentation.

Identification of an *A. millepora* HSF1 Ortholog. A BLASTP search using the human HSF1 protein sequence (National Center for Biotechnology Information: NP_005517.1) as query against the *A. millepora* protein-models database (63) revealed an apparent HSF1 ortholog (*Acropora_millepora_12629*) (SI Appendix, Fig. S1). A tBLASTn search using this *A. millepora* sequence as query against the *A. millepora* genome database (64) identified a single contig (c1011140) containing the entire *HSF1* gene. Alignment of the *A. millepora* *HSF1* mRNA sequence from the transcriptome (65) against the contig c1011140 sequence predicted a gene containing 12 exons spanning ~10 kb (Fig. 1A).

Design and Synthesis of sgRNAs Targeting *HSF1*. The sequence of *HSF1* was analyzed to identify sgRNA target sites that were predicted to be both efficient and unique in the genome using a previously developed pipeline (42–44). Briefly, all potential sgRNA target sites (i.e., sequences that matched the pattern [G, C, or A]N₂₀GG, as well as those for which the reverse-complement matched this pattern) were extracted from the exon sequences. All guides that had exact matches elsewhere in the *A. millepora* genome were identified using Bowtie2 (66) and removed from consideration. Possible sgRNAs with GC content >55% and those containing polyT motifs, which are a terminator for effective transcription of sgRNAs (67), were also removed. Next, possible sgRNA secondary structures were calculated using the Vienna RNA-fold package (68), and sgRNAs for which the stem-loop structure could not fold correctly for Cas9 recognition (69) were removed unless the folding energy was above -18 kcal/mol, where alternative structures would be very unstable. Next, the specificity of each potential sgRNA was evaluated using an optimized reimplement of the Zhang tool (70), which provides a score for the off-target risk, and sgRNAs that scored below 75 were removed. In the rare cases where there were single-nucleotide differences between the mRNA and genomic sequences, we only considered targets that were identical in both sequences. The 45 potential sgRNAs that satisfied all these criteria were considered high quality and suitable for use. We then chose two such sites (one in exon 3 and one in exon 9) for use based on three criteria. (i) They are far enough downstream of the putative start of the coding sequence to avoid complications from alternative transcriptional start sites (if any). (ii) They are far enough upstream of the putative end of the coding sequence that any frame-shifting mutations should severely truncate HSF1. (iii) They have two 5' guanines, because it has been shown that sgRNAs transcribed in vitro by T7 RNA polymerase are most effective at producing mutations in vivo if they start with these nucleotides (71, 72).

To generate DNA templates for the in vitro transcription of each sgRNA, we added the T7 promoter sequence to the 5' end and the sgRNA scaffold sequence to the 3' end of the target-site sequence (SI Appendix, Table S1) and had these templates synthesized using the gBlocks DNA synthesis service (IDT). The templates were reconstituted in pure H₂O to a concentration of 300 nM, and 2 μ L were used in 10- μ L in vitro transcription reactions using the MEGAshortscript T7 Transcription Kit (ThermoFisher no. AM1354). The resulting RNAs were purified using a RNA Clean & Concentrator Kit (Zymo Research no. R1013), and their quality and size were verified using gel electrophoresis and quantified using a NanoDrop2000 Spectrophotometer (ThermoFisher). sgRNA/Cas9 ribonucleo-protein complexes were then generated as described previously (18).

Microinjection and Identification of Successfully Injected Larvae. Microinjection was performed essentially as described previously (18). However, to better control injection volume and aid in the identification of successful injected larvae, we coinjected the sgRNA/Cas9 complexes with a fluorescent dextran indicator (0.2 μ g/ μ L of Alexa Fluor 488-labeled dextran [10,000 MW, Anionic, Fixable]; Invitrogen no. D22910). The injections were performed using a fluorescence microscope (GFP filter, excitation 488 nm) to allow easy monitoring of the injected volume. We attempted to inject a volume equal to ~10% of the total volume of the zygote. At 12 h postfertilization, the fluorescent dextran was still visible in the developing embryos, which were then manually sorted into fluorescent (successfully injected) and nonfluorescent (not successfully injected) groups (SI Appendix, Fig. S2).

Molecular Characterization of the Induced Mutations. At 6 d postfertilization, individual larvae were placed in 100% ethanol and stored at -20 °C until further processing. To extract DNA from individual larvae, the ethanol was replaced with 20 μ L of lysis buffer (10 mM Tris-Cl, pH 8.3, 50 mM KCl, 1.5 mM MgCl₂, 0.3% Tween-20, 0.3% Nonidet P-40), and the larval tissue was disrupted and lysed by crushing with a pipette tip followed by incubation at 94 °C for 20 min. Proteins were degraded by adding 1 μ L of 20 mg/mL Proteinase K (Ambion no. AM2546), incubating for 60 min at 55 °C, and inactivating the proteinase by incubating for 20 min at 94 °C. Aliquots (0.75 μ L) of each such crude extract was used directly in a 20- μ L PCR using the primers described in Fig. 1A and SI Appendix, Table S1, and the following parameters: one cycle of 95 °C for 3 min; 34 cycles of 95 °C for 15 s, 55 °C for 15 s, and 72 °C for 15 s; and one cycle of 72 °C for 5 min. The resulting PCR products were verified using gel electrophoresis, purified with a Qiagen PCR Purification Kit, and cloned using the Topo TA Cloning Kit (Invitrogen no. 451641). To assess the mutational spectra, 10–16 clones for each exon in each larva were Sanger sequenced using the M13 reverse primer (SI Appendix, Table S2).

Assessment of Heat Tolerance. Heat-stress experiments were performed using both uninjected larvae and larvae that had been injected with Cas9 protein alone or with the sgRNA/Cas9 complexes; these experiments began at 3 d postfertilization for Experiment 1 and 4 d postfertilization for Experiments 2 and 3. Individual larvae were placed into separate wells of a 96-well plate containing 200 μ L of filtered seawater per well, and the plates were moved from a 27 °C incubator directly into a 34 °C incubator and held for 48 h. Every 12 h, the larvae were scored as alive if they were intact (whether swimming or immobile) or dead if they had lost structural integrity (Fig. 2A). Control treatments at 27 °C were set up and scored similarly.

Data Availability. All data for this paper are provided in the main text or SI Appendix.

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